

Article

Estimating Biomass in *Eucalyptus globulus* and *Pinus pinaster* Forests Using UAV-Based LiDAR in Central and Northern Portugal

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Abstract

Accurate biomass estimation is important for forest management and climate change mitigation. This study evaluates the potential of using LiDAR (Light Detection and Ranging) data, acquired through Unmanned Aerial Vehicles (UAVs), for estimating above-ground and total biomass in *Eucalyptus globulus* and *Pinus pinaster* stands in central and northern Portugal. The acquired LiDAR point clouds were processed to extract structural metrics such as canopy height, crown area, canopy density, and volume. A multistep variable selection procedure was applied to reduce collinearity and select the most informative predictors. Multiple linear regression (MLR) models were developed and validated using field inventory data. Random Forest (RF) models were also tested for *E. globulus*, enabling a comparative evaluation between parametric and machine learning regression models. The results show that the 25th height percentile, canopy cover density at two meters, and height variance demonstrated an accurate biomass estimation for *E. globulus*, with coefficients of determination (R^2) varying between 0.86 for MLR and 0.90 for RF. Although RF demonstrated a similar predictive performance, MLR presented advantages in terms of interpretability and computational efficiency. For *P. pinaster*, only MLR was applied due to the limited number of field data, yet R^2 exceeded 0.80. Although absolute errors were higher for *Pinus pinaster* due to greater biomass variability, relative performance remained consistent across species. The results demonstrate the feasibility and efficiency of UAV LiDAR point cloud data for stand-level biomass estimation, providing simple and effective models for biomass estimation in these two species.

Keywords: above-ground biomass; point clouds; applied remote sensing; forestry inventory; multiple linear regression; machine learning; dendrometric parameters; geospatial data



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1. Introduction

Forest inventory is a fundamental approach for acquiring data in forested areas. By collecting dendrometric variables from sampled trees, parameters such as wood volume, woody biomass, carbon content, tree dimensions, and spatial distribution can be determined [1]. These data support forest management decisions aligned with predefined objectives. Beyond volumetric assessments, forest inventories can provide information relevant to forest management, biodiversity, carbon cycling, and interactions with climate and soil conditions [2–4]. However, traditional forest inventory methods require extensive fieldwork, involving the measurement of individual trees across multiple plots. This process is time-consuming, requires trained professionals, incurs high costs, and often demands access to remote and complex terrain [5].

Over time, inventory techniques have evolved from visual estimations in the 18th century to modern methods incorporating remote sensing technologies, including satellites, manned aircraft, and unmanned aerial vehicles (UAVs) [6]. These platforms support different types of sensors, depending on their payload capacity [7,8]. In recent years, improvements in spatial resolution have increased the relevance of airborne platforms, particularly due to the development of high-resolution light detection and ranging (LiDAR) systems capable of capturing dense point clouds at low flight speeds [9,10]. LiDAR sensors have proven their effectiveness for capturing data on forest structure and composition [11]. They operate on laser emitters and receivers, inertial navigation systems (INSs), and the Global Navigation Satellite System (GNSS) to ensure accurate capture of multiple return signals, even under dense canopies [12]. The resulting three-dimensional point clouds represent vegetation structure, enabling measurements such as tree height and canopy density [13,14]. Initially deployed to manned aircraft [15], LiDAR sensors are now increasingly used on UAVs, improving the flexibility and efficiency of spatial data acquisition [16]. Despite the high initial acquisition costs and the need for technical expertise [17], LiDAR-based data can be more cost-effective than traditional sampling methods, offering significant financial advantages [18,19]. Additionally, optimizing ground sample sizes allows UAV-LiDAR applications with minimal field effort, while increasing the surveyed area, maintaining accuracy, and reducing costs [20].

LiDAR has been used in forest assessment, demonstrating its capability to estimate, model, and characterize forest structures and ecological parameters [21–23]. LiDAR-based inventory systems support rapid evaluations of forest metrics, including canopy structure, tree height, wood volume, biomass, carbon stocks, and geographic positioning [24,25]. Dominant tree height models generated by LiDAR correlate with field measurements [26]. Moreover, combining UAV-LiDAR data with tools such as dendrochronology techniques and thinning simulators expands its utility for monitoring forest growth and structural dynamics [27]. The use of UAV-LiDAR has demonstrated effectiveness for detecting and measuring individual trees. For instance, Rodríguez-Puerta et al. [28] demonstrated the effectiveness of UAV-based LiDAR data for identifying individual trees and measuring height in young forests. This approach has been further refined by Pirotti [29], who applied template matching to extract tree height and position from LiDAR-derived canopy height models (CHMs) in *Pinus pinaster* Ait. stands. Picos et al. [30] used UAV-LiDAR and CHMs to detect individual trees in *Eucalyptus* plantations, while Solares-Canal et al. [31] explored the use of portable LiDAR for tree identification and segmentation in *Pinus* species. UAV-LiDAR integration has also shown utility in fuel strip planning and management, contributing to fire prevention strategies based on fuel distribution [32,33]. Fernández-Guisuraga et al. [32] highlighted the importance of above-ground biomass, estimated from LiDAR data, spectral, and field inventory data, as a determining factor of fire severity in *P. pinaster* ecosystems. Míguez and Fernández [34] combined normalized difference

vegetation index (NDVI) and high-density LiDAR data to assess the natural regeneration of *P. pinaster* following a high-severity fire disturbance. LiDAR applications also extend to mapping and monitoring forest disturbances such as pests and diseases. Yu et al. [35] proposed a combined approach using UAV hyperspectral and LiDAR data to detect diseases such as pine wilt disease using machine learning algorithms at an individual tree level.

Forest biomass is a key indicator of ecosystem productivity and forest development [36]. Accurate biomass estimation is critical for assessing climate change impacts, particularly in relation to carbon sequestration and storage [37,38]. LiDAR point clouds can be used to estimate biomass based on dendrometric variables [39,40] and are increasingly integrated into forest management strategies, such as monitoring selective logging impacts on biomass and carbon [41]. Biomass estimates from LiDAR data have matched those from traditional inventories in both *Eucalyptus* plantations [42] and complex ecosystems like the Amazon rainforest [18,43]. When validated with field inventories, UAV-LiDAR showed promising results for developing biomass estimation models in tropical and subtropical forests [44,45], showing better accuracy when compared with optical remote sensing techniques [46]. Goldberg et al. [47] used airborne LiDAR data for carbon stock quantification in northern Australia's savannas, focusing on *Eucalyptus miniata* A.Cunn. and *Eucalyptus tetrodonta* F.Muell. Similarly, Pinedo et al. [48] used UAV-LiDAR to estimate tree height and above-ground biomass in *Eucalyptus globulus* Labill. plantations, highlighting the technology's potential for monitoring forest resources. Additionally, Lu et al. [38] explored the estimation of aboveground biomass in *Robinia pseudoacacia* L. forests using point clouds acquired through UAV and backpack LiDAR. Furthermore, Torre-Tojal [49] used random forest algorithms to estimate the above-ground biomass of *Pinus radiata* D.Don based on LiDAR data, demonstrating the versatility of LiDAR technology in forest biomass assessment.

Modeling approaches, including machine learning regressors, have been employed in above-ground biomass (AGB) estimation, often providing more accuracy in comparison with traditional linear models [46,50]. However, stepwise linear regression is still being adopted in the estimation of forest biomass [13,51–53], enabling variable selection to optimize model performance while preserving accuracy [52–54]. Liu et al. [55] reported that machine learning algorithms applied to UAV stereo imagery have achieved higher accuracy in eucalyptus plantations. The integration of LiDAR data with multispectral images and machine learning has shown significant potential. Rodríguez-Puerta et al. [56] compared machine learning algorithms for fuel strip planning in urban-forest interface using a combination of Sentinel-2 with UAV LiDAR and airborne laser scanning (ALS) data. A random forest model was the most reliable, with an overall accuracy of 90.66 per cent in the training set and 91.80 per cent in the test set. Stepwise linear regression has also shown high accuracy in pine forests [57,58]. However, model performance varies by forest type due to species-specific growth and distribution patterns. For example, pine stands often show greater quantitative errors, due to higher biomass variability [59,60]. Plantation age also influences the estimation of parameters derived from LiDAR data [28,34] and the detection of individual trees [61].

Another important factor is the number of metrics that can be derived from UAV-based LiDAR data, many of which correlate with field-based AGB measurements. These metrics include height amplitude, mean canopy height, canopy coverage, and canopy density [62], with mean canopy height showing a strong correlation with AGB [63]. LiDAR-based height percentiles have also proven to be accurate AGB predictors [44,50]. Simplified models based on these parameters show high coefficients of determination (R^2) and relatively low errors, supporting their applicability for biomass estimation. Yan et al. [64] identified mean tree height as the most influential parameter in univariate regression models. In eucalyptus

plantations, tree height and canopy depth were found to be the most effective predictors of trunk volume using a principal components analysis [65]. Leite et al. [66] evaluated both area-based and individual-tree-based methods, incorporating central tendency and distribution metrics. Similarly, Liu et al. [67] extracted multiple LiDAR-derived metrics for *Larix olgensis* A.Henry, including total tree height, crown width, and crown area across height percentiles.

As modeling approaches evolve, their integration with UAV-LiDAR is expected to further improve AGB and total biomass (TB) estimation for different species, serving as a precise and cost-effective tool for carbon monitoring and sustainable forest management [36,68,69]. This is especially relevant in countries with poor forestry management and highly variable and fragmented landscapes, such as in Portugal. In this context, Portuguese forestry is predominantly characterized by a distribution of *P. pinaster* (maritime pine) and *E. globulus* [70]. These species are widespread due to their adaptability to local conditions. However, the terrain across Portugal is variable, with steep slopes, rugged landscapes, and varied soil types presenting a significant management challenge [71]. Forest management has historically been sporadic and uncontrolled, leading to high tree densities, particularly of fast-growing stands of *P. pinaster* and *E. globulus* [72]. High tree density combined with the challenging terrain amplifies the susceptibility of these forests to fire disturbances [73]. The dense undergrowth, along with the flammability of both *P. pinaster* and *E. globulus*, creates an environment where fires can spread rapidly and uncontrollably. These conditions are aggravated by periods of drought and extreme weather, which have become more frequent due to climate change and socioeconomic factors [74,75].

Addressing these issues requires sustainable forest management practices that balance ecological and economic concerns [76]. This study aims to support forest sustainability by exploring the use of UAV-based lidar data to model biomass in *P. pinaster* and *E. globulus* stands. To meet this goal, biomass estimates derived from traditional forest inventory and UAV-based LiDAR data were acquired in the central and northern regions of the country. This allows for the evaluation of the accuracy of high-spatial-resolution remote sensing approaches for improving AGB and TB estimation in these two species under Portuguese forestry conditions.

2. Materials and Methods

2.1. Study Areas

The areas used in this study are distributed across different regions of mainland Portugal (Figure 1). These areas correspond to forest stands of *E. globulus* and *P. pinaster*, spanning from north to south. *E. globulus* stands are located in three areas: Parque das Serras do Porto (Figure 1A) within the municipalities of Paredes and Valongo, in the northern region; Serra da Lousã, within the municipalities of Góis (Figure 1B), in the central region; and in the municipality of Nisa in the Alentejo, southern region (Figure 1C). The *P. pinaster* stands are also located in Serra da Lousã, within the municipalities of Góis and Castanheira de Pêra (Figure 1B) and Ribeira de Pena (Figure 1D), in the north of Portugal.

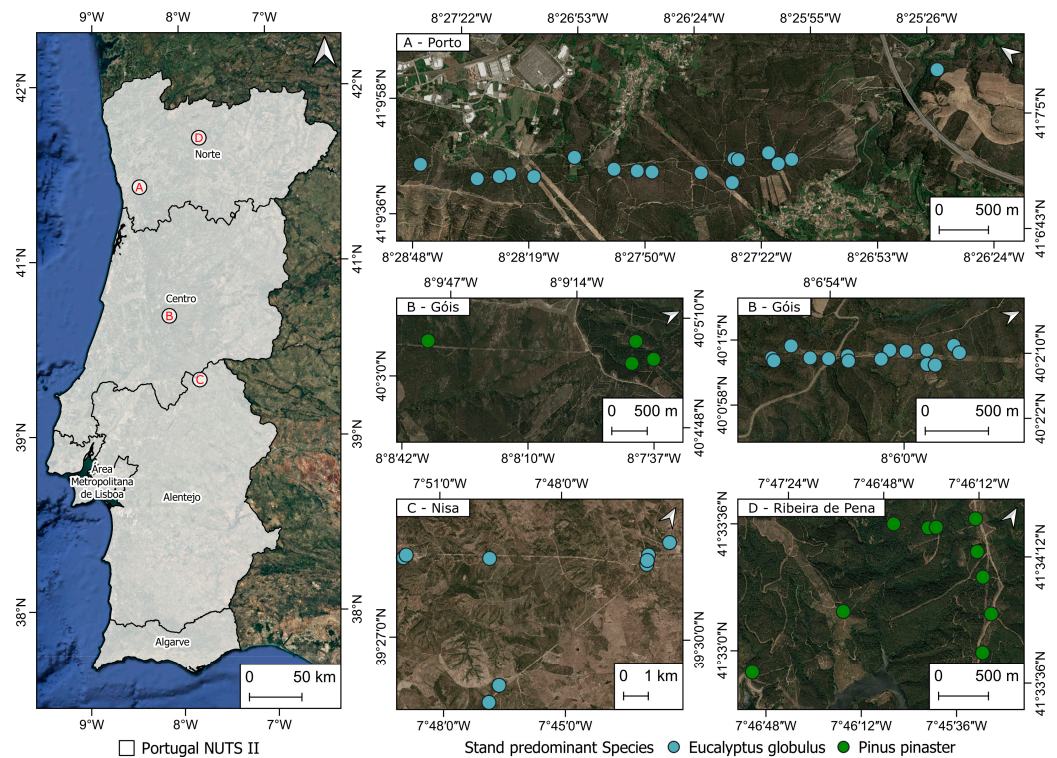


Figure 1. Location of the study areas in mainland Portugal and respective sample stands. Images from Google Earth (accessed on 20 February 2025) and obtained from the National Geographic Information System (SNIG), OrtoSat true color image of mainland Portugal 2023.

2.2. Forest Inventory

The sampling procedure was carried out using two methods: field-based forest inventory and UAV-based LiDAR acquisition. Field data were collected between January 2022 and March 2023. Circular sampling plots of 500 m² were established on a horizontal plane in stands *P. pinaster* and *E. globulus* stands.

Plots on flat terrain (slope < 5°) are not corrected for slope, with a fixed radius of 12.62 m. In sloped areas, inclination was measured with a hypsometer to determine the corrected radius required to maintain a plot area projected on the horizontal plane. This corrected radius is obtained by

$$cr = \frac{r}{\sqrt{\cos \alpha}} \quad (1)$$

where cr is the corrected radius, r the radius at the horizontal plane, and α the terrain inclination (°).

Trees higher than two meters were measured unless the stand was young (50% of trees < 1.3 m), in which all trees were measured. A GNSS receiver was used to record the central coordinates of each sample plot.

In each plot, all trees were measured and identified. The recorded variables included species, diameter at breast height (DBH), total height (H_t), crown base height (H_c), and crown projection diameters in the north–south (CDNS) and also east–west (CDEW).

The dominant height (H_{dom}) is defined as the average height of the 100 trees with the largest DBH per hectare. In a 500 m² sampling plot, this corresponds to an average height of the five trees with the largest DBH. The H_{dom} was calculated for each sampling plot and was used to estimate partial biomass for each tree. Tree-level partial biomass was estimated using established equations based on DBH or DBH and H_{dom} . TB per sampling plot was then extrapolated to a per-hectare basis. Biomass was calculated using equations from the

6th Portuguese National Forest inventory, based on studies developed by Tomé et al. [77], Faias et al. [78], and Soares and Tomé [79], as presented in Table 1.

Table 1. Equations for calculating *Pinus pinaster* and *Eucalyptus globulus* biomass per compartment. AGB: above ground biomass; TB: total biomass; DBH: diameter at breast height; H_t : total tree height.

Compartment	<i>Pinus pinaster</i>	<i>Eucalyptus globulus</i>
Trunk	$w_s = 0.0146dbh^{1.94687}ht^{1.106577}$	$w_s = 0.009964dbh^{1.780459}ht^{1.369618}$, if $hdom > 10.7100$ $w_s = 0.009964dbh^{\beta_2}ht^{1.369618}$, if $hdom \leq 10.7100$ $\beta_2 = \frac{hdom}{-0.70909 + 0.627861hdom}$
Bark	$w_b = 0.0114dbh^{1.8728}ht^{0.6694}$	$w_b = 0.000594dbh^{2.3794759}ht^{1.0849888}$, if $hdom > 18.2691$ $w_b = 0.000594dbh^{\beta_2}ht^{1.0849888}$, if $hdom \leq 18.2691$ $\beta_2 = \frac{hdom}{-0.69951 + 0.45855hdom}$
Branches	$w_{br} = 0.00308dbh^{2.757606} \left(\frac{ht}{dbh} \right)^{-0.39381}$	$w_{br} = 0.095603dbh^{1.674653} \left(\frac{ht}{dbh} \right)^{-0.85073}$
Foliage	$w_l = 0.09980dbh^{1.392518} \left(\frac{ht}{dbh} \right)^{-0.71962}$	—
AGB	$w_a = w_s + w_b + w_{br} + w_l$	$w_a = w_s + w_b + w_{br} + w_l$
Root	$w_r = 0.2756w_a$	$w_r = 0.2487w_a$
TB	$w_t = w_a + w_r$	$w_t = w_a + w_r$

2.3. LiDAR Data Acquisition and Processing

The LIDAR data were acquired by a third party, covering all study areas. Aerial surveys were conducted using UAVs equipped with LIDAR sensors to obtain georeferenced point clouds. In Parque das Serras do Porto (Figure 1A), a Matrice 600 hexacopter UAV (SZ DJI Technology Co., Ltd., Shenzhen, China) with the Velodyne VLP16 (Velodyne Lidar, Inc., San Jose, CA, USA) sensor was used. In the other study areas, the quadcopter Matrice 300 RTK (DJI, Shenzhen, China) equipped with a Mapper 3+ (YellowScan®, Saint-Clément-de-Rivière, France) was used. Flight surveys were conducted at 40 m above the ground level, at an average speed of 5 m s^{−1} with an average point density greater than 50 points per m². Each point cloud was delivered in LAS (LASer format) and divided into multiple files. The data from each area were merged for analysis.

To extract parameters for each stand, the LIDAR point clouds were processed in three stages: (1) pre-processing, which included merging LAS files, removing noise, and clipping sampling plots; (2) topographic normalization, where the ground was segmented, elevation normalized, and points selected according to a height threshold; and (3) parameter extraction, where different metrics are derived from the normalized point clouds. The steps of this methodology are further described in the following subsections.

2.3.1. Point Cloud Pre-Processing

Since the point clouds may contain non-representative points, a point cloud cleaning operation can be performed to remove duplicate or isolated points corresponding to noise. The Statistical Outlier Removal (SOR) filter in CloudCompare software (version 2.12.1) was used for this purpose. This method calculates the average distance from each point to its k nearest neighbors. Points with distances greater than the average plus a certain number of standard deviations were removed. The inputs for the SOR filter were six points, mean distance estimation, and a value of one for the standard deviation multiplier threshold (nSigma). As a result, a new point cloud was generated as the output of the SOR method, where noise was excluded, improving point distribution and reducing the total number of points.

Next, circular plots were extracted from the point clouds, corresponding to the sampled areas. This step was performed in QGIS by buffering the center of each plot to generate

a circumference and clipping the point cloud using the “lasclip” function from LAStools (rapidlasso GmbH, Gilching, Germany). After clipping, additional outliers, such as power lines, were manually removed from some point clouds using CloudCompare.

2.3.2. Terrain Normalization

After clipping and cleaning, topographic normalization was performed. This step adjusts point elevations (Z-axis) to represent height above ground, enabling comparison across plots, as shown in the example presented in Figure 2. To achieve this, ground points must be distinguished from the remaining points. Ground segmentation was performed using the Simple Morphological Filter (SMRF) algorithm [80], which includes creating a surface map with the minimum elevation from the minimum elevation data, segmenting ground elements, and separating ground points from the original point cloud (Figure 2b). This step was applied using MATLAB (R2024a Update 1).

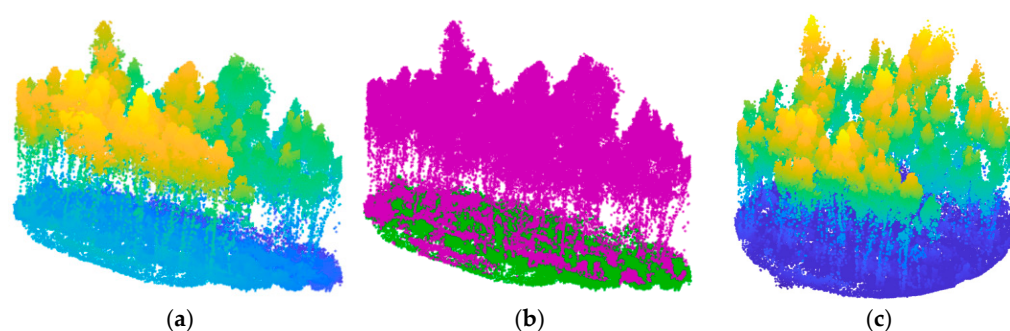


Figure 2. Terrain normalization process applied to a point cloud from one of the surveyed stands: (a) original point cloud; (b) classified ground points (green) and nonground points (purple); (c) normalized point cloud. Colors represent the altitude/height of each point.

The normalization removes the effect of topographic variation on vegetation points. Using the ground points identified from the SMRF algorithm, an interpolation was performed to estimate the ground elevation (Z) for each point. The normalized height was then calculated by subtracting the interpolated ground elevation from each point’s original elevation (Figure 2c). Following point cloud normalization, points below a certain height threshold were removed to exclude vegetation not relevant to tree biomass estimation. This step is carried out to avoid the influence of points that belong to shrub vegetation, which could distort the distribution of the extracted metrics; thus, most points corresponding to tree structures are retained for analysis. Next, a triangulation of the normalized and filtered point clouds was performed to enable volume extraction. This was achieved using the alphaShape algorithm [81], which creates a triangle mesh based on the 3D coordinates (X, Y, Z) of the points and a radius parameter (α) that defines point connectivity; an α of 0.25 was used in this study.

2.3.3. Stand-Level Parameters Extraction

After filtering and normalizing point cloud height, a series of parameters were extracted for each plot analyzed in the scope of this study. These parameters are based on various height percentiles, excluding points below the threshold defined in the previous step (2 m). Three groups of parameters were extracted based on different height percentiles: height, crown structure (density and projected area), and dispersion. A detailed description of the variables for each group of parameters is presented in Table 2.

Table 2. UAV-based LiDAR point cloud metrics derived for each stand.

Variables	Description
$H_{var}, H_{std}, H_{amp}$	Height dispersion metrics: variance, standard deviation, and amplitude (max. – min.)
$H_{P25}, H_{P40}, H_{P50}, H_{mean}, H_{P75}, H_{P90}, H_{P95}, H_{max}$	Height percentiles and statistics at 25th, 40th, 50th (median), mean, 75th, 90th, 95th, and maximum height
$CCD_{min}, CCD_{P25}, CCD_{P40}, CCD_{P50}, CCD_{mean}, CCD_{P75}, CCD_{P90}, CCD_{P95}$	Proportion of area covered by the vertical projection of the tree canopy above a given at different percentiles
$A_{min}, A_{P25}, A_{P40}, A_{P50}, A_{mean}, A_{P75}, A_{P90}, A_{P95}$	Horizontal area covered by canopy at different height percentiles
$V_{min}, V_{P25}, V_{P40}, V_{P50}, V_{mean}, V_{P75}, V_{P90}, V_{P95}$	Estimated canopy volume, computed at different height percentiles

Height parameters are obtained by measuring the 25th, 40th, 50th, 75th, 90th, and 95th percentiles, along with the mean and maximum height, as illustrated in Figure 3. Structure parameters (crown density, projected crown area, and volume) were calculated using points located above 2 m (minimum value) and were computed for each percentile and mean height. Crown density represents the amount of the forest covered by the vertical projection of tree crowns [82], calculated as the proportion of vegetation points relative to the analyzed area. The projected crown area corresponds to the vertical projection of the tree crowns relative to the analyzed plot area and was normalized to hectare units. The volume, also converted to $\text{m}^3 \text{ha}^{-1}$, was estimated from the triangulated surface. Height dispersion was evaluated through variance, standard deviation, and range.

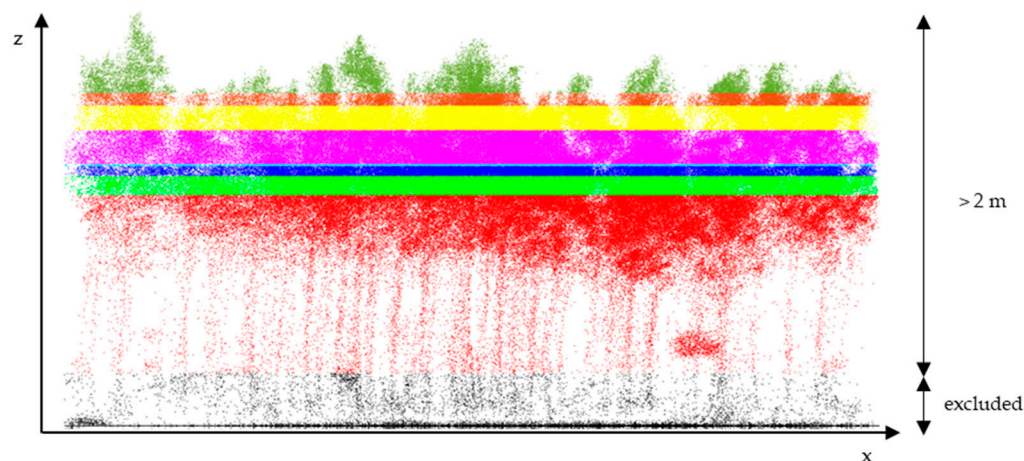


Figure 3. Visual representation of height percentiles and excluded points (below two meters) in a *Eucalyptus globulus* stand. Transition areas between each color represents a height value: black—red, minimum height above two meters; red—green, height at the 25th percentile; green—dark blue, height at the 40th percentile; dark blue—light blue, mean height; light blue—purple, height at the 50th percentile; purple—yellow, height at the 75th percentile; yellow—brown: height at the 90th percentile; brown—green: height at the 95th percentile.

2.4. Data and Statistical Analysis

After extracting geometric and dispersion parameters, plots were grouped by species (*E. globulus* or *P. pinaster*). Each dataset was pre-analyzed to remove plots with insufficient points or inconsistent data. By using the predictor variables based on the extracted parameters (Section 2.3.3), two response variables were modeled, TB and AGB, both expressed in megagrams per hectare (Mg ha^{-1}). The dataset of each species was randomly split

into train (70%) and test (30%) subsets, allowing the validation on data not used during model training.

2.4.1. Feature Selection

The assumptions of normality and homogeneity of variances were tested using the Shapiro–Wilk and Levene’s tests, respectively, in SPSS Statistics 26 (IBM Corp., Armonk, NY, USA) [83]. As most variables violated parametric assumptions ($p < 0.05$), Spearman’s rank correlation coefficient was employed to assess the strength and direction of monotonic associations between predictor and response variables [84]. All correlations were interpreted within a 95% confidence interval.

To address multicollinearity, a known limitation in regression modeling where predictors can show strong interdependencies, a preliminary variable screening was conducted. Multicollinearity inflates the variance of coefficient estimates, reduces model interpretability, and undermines predictive accuracy [85]. Given the large initial set of predictors ($n = 35$) and most extracted parameters based on height percentiles, the likelihood of intercorrelation was high.

The selection process began with Spearman’s rank correlation analysis to identify variables moderately to strongly associated with the response variable. To reduce redundancy among predictors, pairwise correlations above 0.95 were removed. Next, variables with high intercorrelation were screened using the variance inflation factor (VIF). The process begins by fitting a linear model with all input variables and calculating initial VIF values (VIF_1) for each predictor. Then the variable with the highest VIF value is removed if this value exceeds the threshold that was set to five [86]. This elimination triggers a recalculation of VIF values for the remaining variables in the updated model, as the removal of one predictor alters the collinearity structure. This process is repeated in a stepwise manner until all remaining variables demonstrate VIF values below the threshold, obtaining the final VIF values (VIF_2). Subsequently, stepwise regression was applied in the variables selected from VIF_2 using the Akaike information criterion (AIC) as the selection criterion to determine optimal predictor subsets for modeling AGB and TB, for each species separately [87]. This iterative procedure balances model simplicity and predictive performance by adding or removing variables based on their statistical contribution [88]. In this step, a forward selection was implemented by adding one variable at a time.

2.4.2. Above-Ground Biomass and Total Biomass Modeling

Predictive modeling was performed using Multiple Linear Regression (MLR) and Random Forest (RF). The trained regression models were fitted using the selected metrics (Section 2.4.1). MLR is an extension of simple linear regression that incorporates multiple predictor variables into a single multivariate framework, allowing the analysis of complex relationships between the response and explanatory variables [89]. RF is an ensemble learning algorithm that constructs a large number of decision trees during training and combines their outputs to improve predictive accuracy and control overfitting. The method’s strength is based on its use of bootstrap aggregation (bagging) and random feature selection at each split, which improves model robustness and reduces variance, particularly in high-dimensional datasets [90–92].

MLR was implemented and analyzed using the R statistical software (version 4.3.2; R Core Team, 2023) [93]. RF regression models [90] were implemented using the randomForest [94] and caret [95] packages in R, with k-fold cross-validation ($k = 5$) to ensure model generalizability and minimize overfitting. Hyperparameter tuning was performed using a grid search approach, and model training was conducted to identify the optimal configuration. Specifically, the number of trees was set to 500, and the number of variables

randomly selected at each split (mtry) ranged from 2 to 6. Due to the limited number of *P. pinaster* plots ($n = 15$), only MLR was applied as an exploratory approach. Machine learning models such as random forest typically require larger datasets to avoid overfitting and ensure generalizability [96]. Therefore, and in accordance with literature that discourages the use of ML models under such conditions [97], MLR was employed for this dataset, while MLR and RF were both applied for *E. globulus* data.

Model performance was evaluated using three commonly adopted regression metrics: the coefficient of determination (R^2), root mean square error (RMSE), and mean absolute error (MAE). These metrics were selected to provide a complementary understanding into the models' predictive accuracy and robustness, by quantifying the proportion of variance in the dependent variable explained by the predictors for model fit (R^2), and by offering an evaluation of model accuracy (RMSE and MAE), with RMSE penalizing large discrepancies more heavily and MAE providing a straightforward interpretation of average error [98]. These performance metrics were computed for training and test datasets, as well as for the full dataset to support final model evaluation.

3. Results

3.1. Data Characterization and Analysis

Table 3 summarizes the variables and estimated metrics for biomass estimation. Among the two studied species, *P. pinaster* stands showed the highest mean TB (416.3 Mg ha^{-1}), while the lowest TB was observed in an *E. globulus* stand (3.4 Mg ha^{-1}). The lowest TB in a *P. pinaster* stand was 113 Mg ha^{-1} . Maximum TB values were 680.8 Mg ha^{-1} for *P. pinaster* and 107.7 Mg ha^{-1} for *E. globulus*. Similar trends are observed for AGB, with the highest mean value observed in *P. pinaster* stands (403.6 Mg ha^{-1}). The minimum AGB was observed in an *E. globulus* stand (2.7 Mg ha^{-1}), while the minimum in *P. pinaster* was 105.3 Mg ha^{-1} . The maximum AGB values reached 664.2 Mg ha^{-1} for *P. pinaster* and 86.3 Mg ha^{-1} for *E. globulus*. Regarding stand-level canopy structure metrics, *P. pinaster* stands show a higher mean maximum height (18.7 m) compared with *E. globulus* stands (14.9 m). Height distribution in *P. pinaster* stands ranges from 9.7 m to 26 m , while *E. globulus* heights range from 6.6 m to 21.7 m . Crown density also follows this trend, *P. pinaster* stands have a mean density of 37.6% , ranging between 2.9% and 67.9% , while *E. globulus* stands have a mean density of 20.7% , reaching a maximum of 45.2% and a minimum of 1.1% . The crown projection area is higher in *P. pinaster* stands ($5059.2 \text{ m}^2 \text{ ha}^{-1}$) when compared with the *E. globulus* stands ($3318.1 \text{ m}^2 \text{ ha}^{-1}$). Volume values show smaller differences between species. The mean volume is $3921.9 \text{ m}^3 \text{ ha}^{-1}$ in *P. pinaster* stands and $3198.5 \text{ m}^3 \text{ ha}^{-1}$ in *E. globulus* stands. The minimum and maximum values were both observed in *E. globulus* stands, ranging from $7 \text{ m}^3 \text{ ha}^{-1}$ to $13,602.8 \text{ m}^3 \text{ ha}^{-1}$.

Table 3. Distribution of metrics for the evaluated *Eucalyptus globulus* and *Pinus pinaster* stands. SD: standard deviation; P25: value at the 25th percentile; P75: value at the 75th percentile; TB: total biomass; AGB: above ground biomass.

Parameter	Species	Mean $\pm$ SD	Min.	P25	Median	P75	Max.
TB (Mg ha^{-1})	<i>E. globulus</i>	42.14 ± 25.21	3.41	24.36	40.09	59.83	107.72
	<i>P. pinaster</i>	416.30 ± 189.23	112.97	289.10	452.01	576.60	680.82
AGB (Mg ha^{-1})	<i>E. globulus</i>	33.74 ± 20.19	2.73	19.51	32.10	47.91	86.26
	<i>P. pinaster</i>	403.62 ± 185.79	105.34	279.57	440.67	561.46	664.17

Table 3. Cont.

Parameter	Species	Mean $\pm$ SD	Min.	P25	Median	P75	Max.
Max. height (m)	<i>E. globulus</i>	14.93 $\pm$ 3.53	6.61	12.47	14.48	17.35	21.65
	<i>P. pinaster</i>	18.69 $\pm$ 4.78	9.66	15.90	18.66	22.66	25.99
Canopy density (%)	<i>E. globulus</i>	20.71 $\pm$ 10.14	1.09	12.55	19.80	28.74	45.19
	<i>P. pinaster</i>	37.62 $\pm$ 19.18	2.86	27.54	36.33	47.14	67.85
Crown area (m ² ha ⁻¹)	<i>E. globulus</i>	3318.07 $\pm$ 1493.64	139.67	2336.56	3305.69	4293.84	6672.64
	<i>P. pinaster</i>	5059.16 $\pm$ 2554.35	328.89	3907.07	4605.55	6639.56	9444.33
Volume (m ³ ha ⁻¹)	<i>E. globulus</i>	3198.48 $\pm$ 2987.50	7.03	1429.90	1927.56	4415.91	13,602.75
	<i>P. pinaster</i>	3921.92 $\pm$ 3009.54	53.66	1412.93	3630.60	4480.60	11,199.45

The correlation between each individual metric estimated from the LiDAR point cloud data with AGB and TB is presented in Table 4. In *E. globulus* stands, correlation coefficients for height metrics range from $r = 0.85$ for maximum height to $r = 0.95$ for mean height. Among dispersion metrics, height amplitude shows the highest correlation ($r = 0.86$), while height variance and standard deviation were similar ($r = 0.54$). Canopy cover density (CCD) at the 90th percentile had the highest correlation for this metric type ($r = 0.61$), while density at the minimum height above 2 m showed the weakest correlation ($r = 0.51$). For the crown projection area, the highest correlation was achieved when considering points above 2 m ($r = 0.56$). Volume-related metrics showed the weakest correlations overall, with values ranging from 0.30 for minimum height to 0.51 at the 95th height percentile. As for *P. pinaster* stands (Table 4), height appears as the most strongly correlated parameter, ranging from $r = 0.90$ and 0.92 (maximum height in AGB and TB, respectively) to $r = 0.94$ and 0.96 (mean height and height at 40th and 50th percentiles). Among dispersion metrics, amplitude showed a high correlation ($r = 0.92$ for AGB and $r = 0.90$ for TB), while variance and standard deviation were below 0.5. For canopy density, the highest correlation was observed at the minimum height (2 m) with $r = 0.52$ for AGB and $r = 0.53$ for TB, whereas the lowest was at the 95th percentile ($r = 0.16$ and 0.20). Crown area above 2 m showed the strongest correlation in this category, while crown area at the 90th and 95th percentiles was below 0.3. As for volume metrics, the weakest was observed for volume at the 90th percentile, while the mean volume showed the highest correlation values in this type of metric.

Table 4. Correlation coefficients (r) and variance inflation factors (VIF) for predictor variables of aboveground biomass (AGB) and total biomass (TB) in *Eucalyptus globulus* and *Pinus pinaster*. VIF values are identical for AGB and TB. H: height; A: projected crown area per hectare (m² ha⁻¹); V: volume per hectare (m³ ha⁻¹); CCD: canopy cover density.

Variables	<i>Eucalyptus globulus</i>				<i>Pinus pinaster</i>			
	AGB	TB	Both		AGB	TB	Both	
	r	r	VIF ₁	VIF ₂	r	r	VIF ₁	VIF ₂
H _{var}	0.542	0.542	3.17	1.20	0.430	0.333	3.56	1.62
H _{std}	0.542	0.542	—	—	0.430	0.333	—	—
H _{amp}	0.855	0.855	—	—	0.915	0.903	—	—
H _{P25}	0.925	0.925	6.62	1.68	0.927	0.952	—	—
H _{P40}	0.943	0.943	—	—	0.939	0.964	—	—

Table 4. Cont.

Variables	<i>Eucalyptus globulus</i>				<i>Pinus pinaster</i>			
	AGB	TB	Both		AGB	TB	Both	
	<i>r</i>	<i>r</i>	VIF ₁	VIF ₂	<i>r</i>	<i>r</i>	VIF ₁	VIF ₂
H _{P50}	0.944	0.944	—	—	0.939	0.964	—	—
H _{mean}	0.949	0.949	—	—	0.939	0.964	—	—
H _{P75}	0.937	0.937	—	—	0.927	0.939	—	—
H _{P90}	0.924	0.924	—	—	0.927	0.939	—	—
H _{P95}	0.915	0.915	—	—	0.927	0.939	—	—
H _{max}	0.853	0.853	8.49	—	0.903	0.915	2.55	1.75
CCD _{min}	0.513	0.513	13.67	4.71	0.515	0.527	126.35	—
CCD _{P25}	0.548	0.548	—	—	0.455	0.491	—	—
CCD _{P40}	0.551	0.551	—	—	0.394	0.442	—	—
CCD _{P50}	0.559	0.559	—	—	0.394	0.442	—	—
CCD _{mean}	0.581	0.581	—	—	0.455	0.491	—	—
CCD _{P75}	0.571	0.571	—	—	0.394	0.442	—	—
CCD _{P90}	0.605	0.605	66.70	—	0.309	0.345	—	—
CCD _{P95}	0.578	0.578	38.44	3.02	0.164	0.200	—	—
A _{min}	0.564	0.564	—	—	0.430	0.455	219.27	—
A _{P25}	0.521	0.521	—	—	0.358	0.370	—	—
A _{P40}	0.505	0.505	—	—	0.345	0.358	—	—
A _{P50}	0.510	0.510	—	—	0.345	0.358	—	—
A _{mean}	0.501	0.501	—	—	0.430	0.442	—	—
A _{P75}	0.449	0.449	—	—	0.345	0.358	—	—
A _{P90}	0.450	0.450	—	—	0.261	0.273	—	—
A _{P95}	0.437	0.437	18.65	—	0.261	0.273	20.69	—
V _{min}	0.302	0.302	—	—	0.503	0.442	—	—
V _{P25}	0.373	0.373	—	—	0.455	0.418	14.76	1.52
V _{P40}	0.372	0.372	—	—	0.418	0.382	—	—
V _{P50}	0.384	0.384	15.18	2.59	0.418	0.382	—	—
V _{mean}	0.395	0.395	—	—	0.527	0.491	—	—
V _{P75}	0.413	0.413	—	—	0.370	0.345	—	—
V _{P90}	0.459	0.459	—	—	0.321	0.309	—	—
V _{P95}	0.505	0.505	—	—	0.430	0.418	—	—

After filtering for multicollinearity, several variables were excluded using Spearman's correlation values. The variables that were used in the VIF analysis varied across the two species (Table 4). For *E. globulus*, the initial set of variables used were height variance, the 25th percentile of height, maximum height, CCD at the minimum height above two meters and at the 90th and 95th height percentiles, crown area at the 95th percentile of height, and volume at the median height. After the VIF filtering procedure, maximum height and CCD at the 90th percentile of height were excluded. As for *P. pinaster*, the height variance, maximum height, CCD and projected crown area at the minimum height, crown area at the 95th percentile of height, and volume at the 25th percentile of height were used in the VIF procedure, with the final set of variables being composed by height variance, maximum height, and volume at the 25th percentile of height.

3.2. Above Ground and Total Biomass Estimation

Regression models for AGB and TB estimation were fitted using the selected metrics extracted from the LiDAR point clouds of the plots of each species. The results obtained during training were satisfactory, with the models explaining at least 89% of the variance in the dependent variables. The best performance was observed for the *E. globulus* stands, while the highest estimation errors occurred for *P. pinaster*.

3.2.1. Biomass Estimation of *Eucalyptus globulus* Stands

The regression models for *E. globulus* were developed using stepwise linear regression and are presented in (2) and (3). After applying MLR, the best predictive performance for both AGB and TB at the stand level was achieved using the height at 25th percentile (H_{P25}), CCD at two meters height (CCD_{min}), and height variance (H_{var}).

$$AGB = -24.94 + 5.46 \cdot H_{P25} + 1.59 \cdot H_{var} + 24.21 \cdot CCD_{min} \quad (2)$$

$$TB = -31.14 + 6.82 \cdot H_{P25} + 1.99 \cdot H_{var} + 30.23 \cdot CCD_{min} \quad (3)$$

The final models provided an R^2 of 0.89 for both AGB and TB, with an RMSE of 6.7 Mg ha^{-1} and 8.37 Mg ha^{-1} and a MAE of 5.36 Mg ha^{-1} and 6.69 Mg ha^{-1} , respectively (Table 5).

Table 5. Regression performance metrics for above-ground biomass (AGB) and total biomass (TB) in *Eucalyptus globulus* stands. MLR: multiple linear regression; RF: random forest; R^2 : coefficient of determination; RMSE: root mean square error; MAE: mean absolute error.

Model	Dataset	Parameter	AGB	TB
MLR	Train	R^2	0.89	0.89
		RMSE (Mg ha^{-1})	6.70	8.37
		MAE (Mg ha^{-1})	5.36	6.69
	Test	R^2	0.73	0.73
		RMSE (Mg ha^{-1})	10.49	13.10
		MAE (Mg ha^{-1})	8.03	10.03
RF	Train	R^2	0.96	0.96
		RMSE (Mg ha^{-1})	4.69	5.71
		MAE (Mg ha^{-1})	3.61	4.43
	Test	R^2	0.72	0.73
		RMSE (Mg ha^{-1})	10.42	12.92
		MAE (Mg ha^{-1})	7.95	9.76

When applied to the test dataset ($n = 12$), the model performance decreased slightly but remained robust, with $R^2 = 0.73$, RMSE = 10.49 Mg ha^{-1} and 13.1 Mg ha^{-1} , and MAE = 8.03 Mg ha^{-1} and 10.03 Mg ha^{-1} for AGB and TB, respectively. When the model was applied to all *E. globulus* plots, the overall performance was $R^2 = 0.85$, RMSE = 7.9 Mg ha^{-1} and 9.9 Mg ha^{-1} , and MAE = 6.1 Mg ha^{-1} and 7.6 Mg ha^{-1} , for AGB and TB, respectively. The predicted values showed a strong correlation with those derived from the field inventory (Figure 4), with estimations ranging from 0.0 to 28.5 Mg ha^{-1} for TB (Figure 4c) and from 0.0 to 22.8 Mg ha^{-1} for AGB (Figure 4d) and 18.8 Mg ha^{-1} .

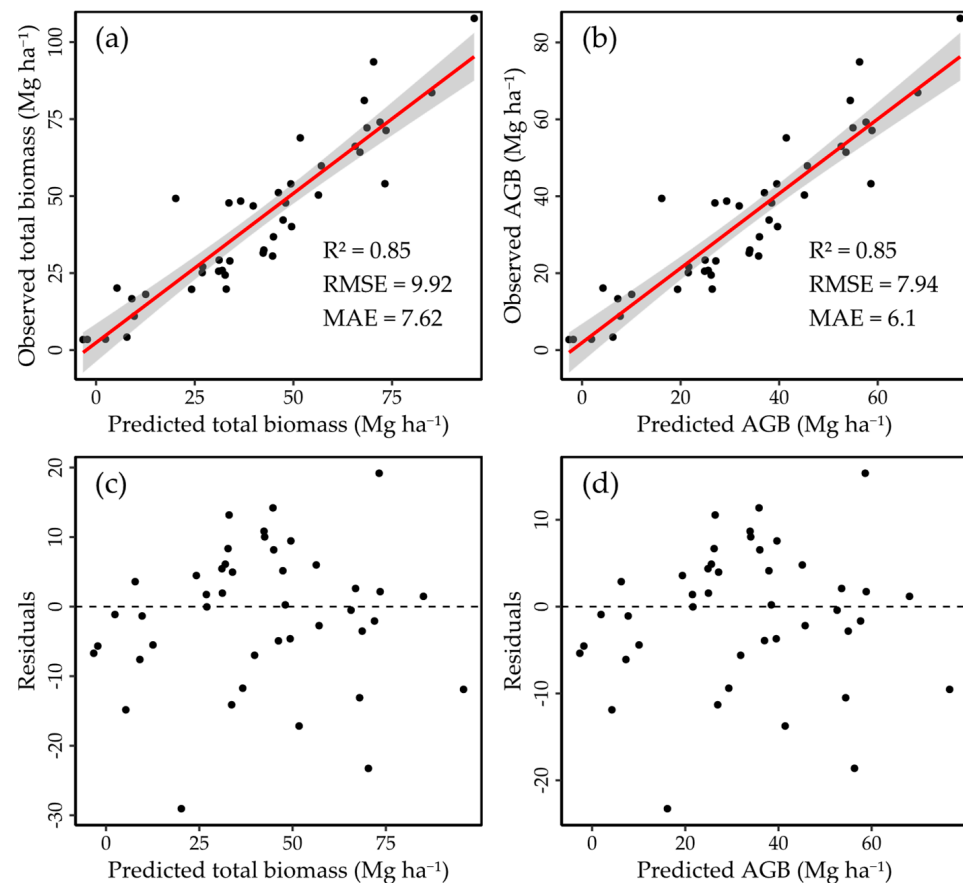


Figure 4. Observed and predicted total biomass (a,b) above ground biomass (AGB) and their respective residual plots (c,d) for *Eucalyptus globulus* data using multiple linear regression.

When applying RF, H_{P25} demonstrates to be the dominant predictor with feature importance of 60.4% and 62.7% for TB and AGB, respectively, followed by CCD_{min} (20.6% and 19.0%) and H_{var} (19.0% and 18.3%). The RF models showed a strong predictive capability for both TB and AGB at the stand level (Table 5). For AGB, the model achieved an R^2 of 0.96 on the training set, with an RMSE of 4.69 Mg ha⁻¹ and a MAE of 3.61 Mg ha⁻¹. For TB, the training performance was similarly high ($R^2 = 0.96$; RMSE = 5.71 Mg ha⁻¹; MAE = 4.43 Mg ha⁻¹). When tested on independent data ($n = 12$), performance decreased but remained robust, with R^2 values of 0.72 and 0.73, RMSEs of 10.42 Mg ha⁻¹ and 12.92 Mg ha⁻¹, and MAEs of 7.95 Mg ha⁻¹ and 9.76 Mg ha⁻¹ for AGB and TB, respectively. On the full dataset, the overall performance resulted in R^2 values of 0.90 for AGB and TB, with RMSE of 6.53 and 8.25 Mg ha⁻¹ and MAE of 4.76 and 5.71 Mg ha⁻¹, respectively. The predicted values were strongly correlated with measured biomass (Figure 5), with residuals ranging from 0.0 to 28.2 Mg ha⁻¹ for TB (Figure 5c) and from 0.3 to 21.8 Mg ha⁻¹ for AGB (Figure 5d).

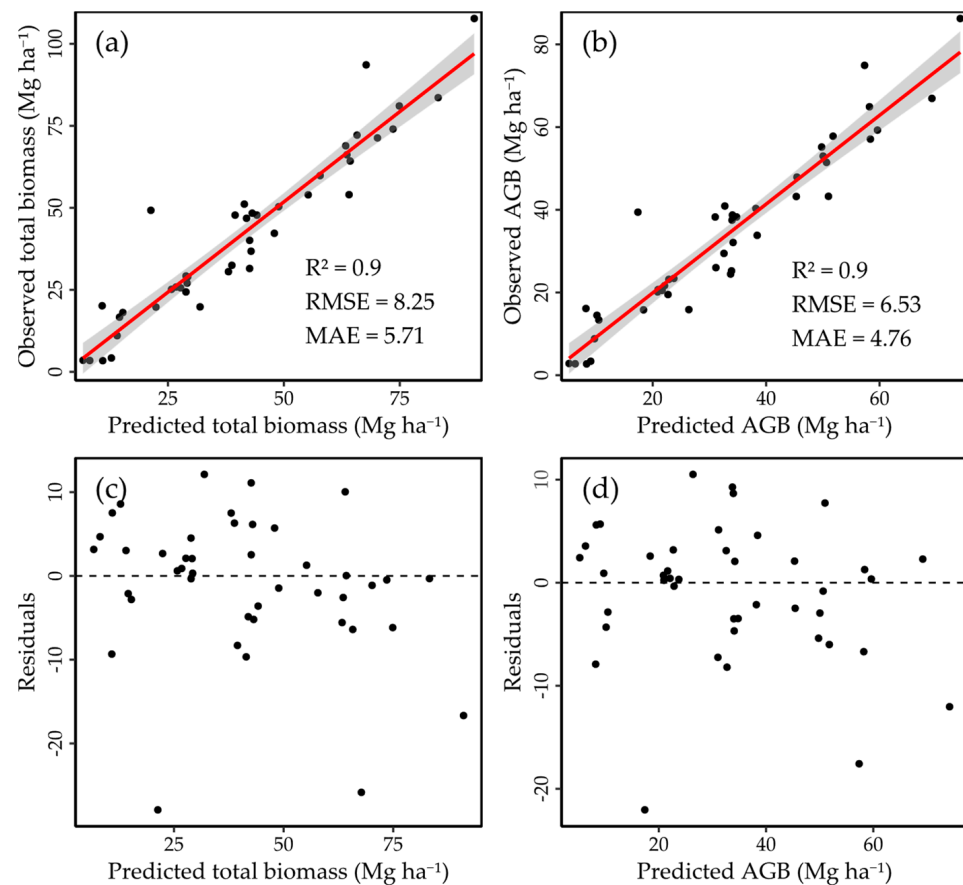


Figure 5. Observed and predicted total biomass (a,b) above ground biomass (AGB) and their respective residual plots (c,d) for *Eucalyptus globulus* data using a random forest.

3.2.2. Biomass Estimation of *Pinus pinaster* Stands

The regression models for *P. pinaster* are shown in (4) and (5) for AGB and TB, respectively. The initial variable selection included height amplitude (above 2 m), mean height, crown area, and canopy density at 2 m. The variable selection included the maximum height ($H_{\max}$) and stand-level volume per hectare at the 25th percentile of height (V_{P25}) as the most effective predictors for both AGB and TB in *P. pinaster*.

$$\text{AGB} = -299.6 + 33.65 \cdot H_{\max} + 0.0172 \cdot V_{P25} \quad (4)$$

$$\text{TB} = -297.7 + 34.12 \cdot H_{\max} + 0.0176 \cdot V_{P25}, \quad (5)$$

The AGB model obtained an R^2 of 0.91, RMSE = 56.43 Mg ha⁻¹, and MAE = 51.8 Mg ha⁻¹. The TB model achieved an R^2 of 0.91, RMSE = 58.53 Mg ha⁻¹, and MAE = 53.31 Mg ha⁻¹. The performance in the test set was similar for R^2 (0.91) and slightly lower for the other metrics, with RMSE = 89.1 Mg ha⁻¹ and 91.59 Mg ha⁻¹ and MAE = 79.37 Mg ha⁻¹ and 81.33 Mg ha⁻¹ for AGB and TB, respectively. These results are presented in Table 6.

When applied to all *P. pinaster* plots, the models achieved $R^2 = 0.88$ for AGB (RMSE = 69.06 Mg ha⁻¹, MAE = 60.99 Mg ha⁻¹) and $R^2 = 0.88$ for TB (RMSE = 71.27 Mg ha⁻¹, MAE = 62.65 Mg ha⁻¹). However, when compared with *E. globulus* (Figures 4 and 5), the estimation errors were higher, ranging from 12.5 to 140.1 Mg ha⁻¹ for TB (Figure 6c) and from 12.5 to 138.4 Mg ha⁻¹ for AGB (Figure 6d).

Table 6. Regression performance metrics for above-ground biomass (AGB) and total biomass (TB) in *Pinus pinaster* stands. R^2 : coefficient of determination; RMSE: root mean square error; MAE: mean absolute error.

Dataset	Parameter	AGB	TB
Train	R^2	0.91	0.91
	RMSE (Mg ha^{-1})	56.43	58.53
	MAE (Mg ha^{-1})	51.80	53.31
Test	R^2	0.91	0.91
	RMSE (Mg ha^{-1})	89.10	91.59
	MAE (Mg ha^{-1})	79.37	81.33

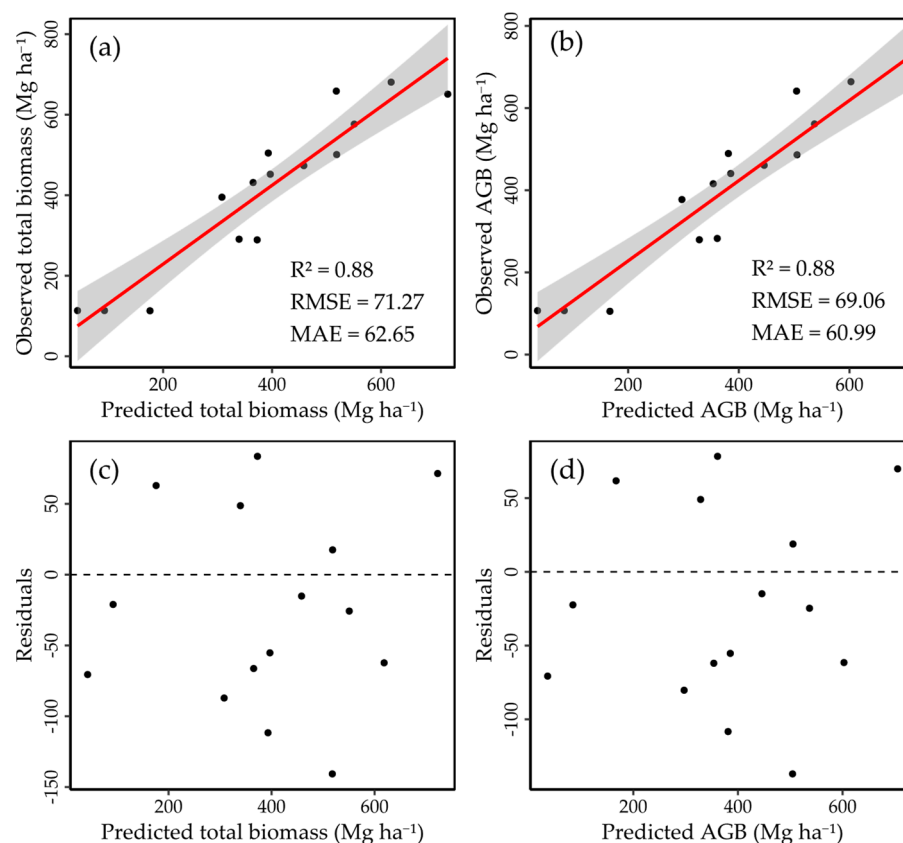


Figure 6. Observed and predicted total biomass (a,b) above ground biomass (AGB) and their respective residual plots (c,d) for *Pinus pinaster* data using multiple linear regression.

4. Discussion

Accurate biomass estimation is an important component of forest monitoring, directly contributing to carbon accounting and sustainable forest management. In this study, MLR and RF models were developed using UAV-based LiDAR data to estimate AGB and TB in *E. globulus*, while only MLR was applied to *P. pinaster* due to limited sample size. MLR is a widely used parametric approach that allows for the selection of the most relevant predictors, which enables interpretable models, efficient computation, and reduced model complexity, but it requires linear relationships and careful multicollinearity control [54]. In contrast, machine learning models such as RF can capture non-linear interactions and tolerate correlated variables but often lack interpretability and require larger datasets. Both of these models are widely used in remote sensing applications for forest biomass estimation [52,53].

To reduce the complexity of the model and improve the accuracy of predictions, a minimal number of parameters needs to be selected through feature selection techniques [99]. The complexity of this task is amplified when there is a high degree of correlation between the predictors, as in the case of this study (Table 4). Therefore, a multistep feature selection process was implemented, which specifically includes Spearman correlation filtering, VIF, and stepwise regression. This filtering approach reduced the initial set of variables to three to four predictors, explaining over 89% of the variance in the training dataset. Such variable selection is essential in remote sensing, where metrics often show high collinearity.

For *E. globulus*, both MLR and RF performed well. MLR achieved $R^2 = 0.89$ in training and $R^2 = 0.73$ in testing (Table 5), with RMSE ranging from 6.70 to 13.10 Mg ha⁻¹. RF obtained a better performance in training when compared with MLR ($R^2 = 0.96$), but both models showed comparable results in the test set ($R^2 = 0.72$ for AGB using RF and $R^2 = 0.73$ for the cases). The reduction from training to test sets is in line with the anticipated model behavior in terms of generalization, as biomass variation is typically influenced by factors specific to the location, environmental heterogeneity, and stand history [100]. On the full dataset (Figures 4 and 5), RF achieved $R^2 = 0.90$ with RMSE of 6.53–8.25 Mg ha⁻¹, while MLR achieved $R^2 = 0.86$ with RMSE of 7.5–9.37 Mg ha⁻¹. These results confirm that canopy height-based metrics extracted from UAV-LiDAR are reliable predictors of biomass in eucalyptus plantations.

The MLR model performed well in *P. pinaster* with $R^2 = 0.91$ in both training and testing (Table 6), but absolute errors were higher (RMSE up to 91.59 Mg ha⁻¹ for TB in the test set), which may be related to the wider biomass range (113–681 Mg ha⁻¹). On the full dataset (Figure 6), the R^2 performance was high at 0.88, with an RMSE of 69.06–71.27 Mg ha⁻¹. Despite the differences in the absolute error, relative errors in the training set between species were comparable (14–17%), indicating a good model generalization across species with different structural complexity [59,60]. The exclusive use of MLR for *P. pinaster* is justified by the limited number of field samples, a common limitation in forest biomass studies [41,43]. It is an established fact that machine learning (ML) typically requires large datasets to reliably capture non-linear interactions [101]. However, it must be noted that this is not always feasible due to the cost and logistical complexity of field sampling [18]. Consequently, MLR was determined to be a more suitable approach to be reported, highlighting the technical merits of field data collection. As observed in Liu et al. [102], even when the number of predictors is reduced, MLR can produce reliable outputs if linear relationships are evident.

In regard to selected model variables, in *E. globulus*, the most significant predictors were identified as H_{P25} , CCD_{min} , and H_{var} . These features characterize vertical structure and canopy heterogeneity, which are ecologically relevant indicators of biomass accumulation. Lower height percentiles, such as H_{P25} , are capable of capturing understory dynamics, while CCD_{min} and H_{var} provide information on foliage distribution and spatial irregularity. For *P. pinaster*, H_{max} and V_{P25} were identified as suitable predictors, highlighting the influence of upper canopy development and low-end volume distribution in biomass differentiation. These results are in line with Domingo et al. [58], who used low-density LiDAR data to estimate TB in *Pinus halepensis*, achieving comparable results ($R^2 = 0.87$) when using, among others, elevation variance and H_{P25} . Yan et al. [64] and Pinedo et al. [48] found that mean canopy height was a dominant predictor for AGB in *Eucalyptus*, achieving R^2 of 0.87 and 0.68, respectively, for AGB estimation. This study obtained similar or better performance, demonstrating the relevance of vertical structure metrics in biomass estimation.

When compared with other remote sensing approaches, the models in this study generally obtained a higher accuracy. Fernández-Guisuraga et al. [32] reported R^2 values of

0.72 and 0.68 for TB and AGB in *P. pinaster* using ALS and Landsat data, while Li et al. [53] achieved $R^2 = 0.41$ using Landsat 8 imagery in a subtropical pine forest. Maesano et al. [103] developed a stepwise regression model combining UAV-RGB and LiDAR data in *Pinus nigra*, achieving $R^2 = 0.81$ and $RMSE = 45.5 \text{ Mg ha}^{-1}$, a comparable R^2 but with lower error than our *P. pinaster* results, which were impacted by greater biomass variability.

While RF showed better performance than MLR in terms of overall fit, the difference in predictive performance on test data was small. Nevertheless, the consistency in performance across datasets is indicative of the reliability of the UAV-based point cloud canopy metrics used for biomass estimation at the stand level. Despite its linear assumptions and simpler structure [101], the MLR model demonstrated a performance comparable to the complex RF model [90]. Other studies demonstrated similar findings with parametric models to match or surpass machine learning techniques, provided that multicollinearity is effectively mitigated and variable selection is conducted with rigor [104–106]. An important strength of the MLR models is their simplicity, especially where decision-makers require interpretability and reproducibility [58]. The linear nature of MLR models facilitates the interpretation of the individual impact of each explanatory variable on biomass estimates, which is advantageous for operational applications. For instance, in *E. globulus*, the positive coefficients for HP25 and CCDmin suggest that both mid-canopy height and lower canopy density are positively related to biomass accumulation, which aligns with ecological expectations. Similarly, for *P. pinaster*, Hmax and VP25 provide interpretable associations to canopy dominance and volume distribution within plots. These relationships are transparent and reproducible, offering information for forest managers and supporting decision-making in resource planning. In contrast, Random Forest-based models, on the other hand, function as “black boxes”, making it difficult to explicitly analyze the role of each variable in the final prediction, which may limit their acceptance in contexts where interpretability is essential, with higher computational cost [107]. Although RF achieved slightly better accuracy, its lack of transparency can limit its adoption in operational forestry. In future work, model-agnostic interpretability tools such as SHAP (SHapley Additive exPlanations) values [108] could be integrated to improve the transparency of RF models and better interconnect variable influence to stakeholders.

One limitation of both modeling approaches is their current restriction to stand-level biomass estimation. However, with appropriate data density and tree segmentation, the methodology could be adapted for individual tree biomass estimation, enabling a refinement of forest inventories, supporting species-level management, and improving biomass mapping in mixed and heterogeneous landscapes. Moreover, the use of other data sources should be considered for data fusion. Li et al. [36] used multispectral and LiDAR data to compare the performance of multiple stepwise regression, random forest, support vector machine, and decision tree algorithms for *E. globulus* biomass estimation. The random forest achieved the best performance with $R^2 = 0.94$ (training) and $R^2 = 0.87$ (test) and lower RMSE values. Lu et al. [38] observed a similar result when combining UAV and backpack LiDAR point clouds in an *R. pseudoacacia* forest, achieving a higher accuracy using random forest when compared with multiple regression ($R^2 > 0.90$). Despite the potential of advanced methods, the simplicity of the MLR models presented in this study provides several practical advantages. These models can be applied to large areas, require lower computational requirements when compared with machine learning approaches, reduce fieldwork, and enable cost-effective, non-destructive biomass monitoring. Moreover, the performance was equally satisfactory when all the available data were used (Figures 4 and 6), suggesting that these models maintain their generalization capacity [109].

5. Conclusions

This study demonstrated the application of UAV-based LiDAR point clouds for estimating above-ground and total biomass in *E. globulus* and *P. pinaster* stands in central and northern Portugal. The resulting models performed well, with high coefficients of determination and relatively low estimation errors, even when applied to independent validation datasets. The simplicity of the final models, requiring few predictor variables, highlights the practical utility of UAV-LiDAR data in forest biomass estimation. The methodology presented offers an efficient, cost-effective, and non-destructive alternative for large-scale forest monitoring, with potential for operational application. Despite higher absolute errors in *P. pinaster* plots, the relative performance was consistent across species, suggesting good generalizability of the methodology. The methodology applied in this study can serve as a non-destructive forest biomass assessment approach, with applications in forest inventories, carbon stock monitoring, and landscape-scale forest management planning. Moreover, this methodology can be evaluated in other species with economic value and different canopy structures, such as *Quercus suber*.

Future research should focus on expanding the dataset to include a greater number and diversity of plots, stratified by species, age class, and stand structure. This would improve model calibration and support more generalizable biomass estimates. The methodology used in this study can also be adapted, with additional data and segmentation techniques, to estimate biomass at the individual tree level. Integrating tree-level structural metrics from UAV-based LiDAR data with species classification derived from multispectral or hyperspectral data would support the development of species-specific, tree-level biomass models. These approaches could be particularly useful in mixed or structurally complex stands. Furthermore, combining structural and spectral information has the potential to improve biomass predictions by capturing complementary information related to canopy architecture, foliage distribution, and physiological traits. Future studies should also prioritize the interpretability of machine learning models, especially when they are used in applied forestry contexts. This will allow practitioners to better understand the underlying ecological drivers of biomass variability and improve trust in model-based decision-making.

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